2024 International Conference on Artificial Intelligence and Quantum Computation-Based Sensor Application (ICAIQSA)

Generative AI-Driven Chatbot for Personalized University Information and Assistance

Deepali Joshi
Information Technology
Vishwakarma Institute of Technology
Pune
deepali.joshi@vit.edu

Shreyash Tekade
Information Technology
Vishwakarma Institute of Technology
Pune
shreyash.tekade21@vit.edu

Sayee Zanzane
Information Technology
Vishwakarma Institute of Technology
Pune
sayee.zanzane211@vit.edu

Dhawal Sakharwade
Information Technology
Vishwakarma Institute of Technology
Pune
dhawal.sakharwade21@vit.edu

Ankur Tripathi
Information Technology
Vishwakarma Institute of Technology
Pune
ankur.tripathi21@vit.edu

Shivam Tiwadi
Information Technology
Vishwakarma Institute of Technology
Pune
shivam.tiwadi21@vit.edu

2024 International Conference on Artificial Intelligence and Quantum Computation-Based Sensor Application (ICAIQSA) | 979-8-3315-1795-3/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICAIQSA64000.2024.10882421

***Abstract*—This paper outlines the development of a generative AI-driven chatbot for university admissions and general inquiries, using a Retrieval-Augmented Generation (RAG) framework. By embedding university-related documents into a vector database and indexing them with FAISS, the system efficiently retrieves information and generates precise, context-aware responses. The chatbot improves user experience by providing accurate answers to prospective students and other users, showcasing the potential of generative AI in educational settings for managing inquiries effectively**

Keywords—Generative AI, Retrieval-Augmented Generation (RAG,), University Chatbot, Natural Language Processing (NLP), Information Retrieval.

I. Introduction

The demand for accessible and accurate information in educational institutions is growing, with prospective students seeking quick answers about admissions, programs, and campus facilities. Traditional methods of handling these inquiries often fall short. To address this, we propose a generative AI-driven chatbot designed to assist with university admissions and general inquiries.

Our chatbot utilizes a Retrieval-Augmented Generation (RAG) framework, which integrates retrieval and generation techniques to deliver precise and contextually relevant responses. This involves embedding university-related documents into a vector space and indexing them in a vector database. Upon receiving a query, the system retrieves relevant documents and generates accurate responses.

The primary goal is to enhance the user experience by providing accurate and timely information. By leveraging the RAG framework, we aim to improve the efficiency and relevance of the chatbot's responses. This paper outlines the methodology, including the creation of vector embeddings, indexing, and the implementation of a conversational chain. We also discuss the technologies used and present the results, demonstrating the effectiveness of the RAG framework in providing context-aware responses.

This research highlights the potential of generative AI to transform information retrieval in educational settings, offering a scalable solution for managing inquiries accurately and efficiently.

II. Literature Review

The paper addresses the scarcity of scholarly literature on the use cases of AI chatbots in Applied English Language Studies (AELS) within the Higher Education (HE) sector, emphasizing the novelty of exploring this area. It mentions the introduction of GPT-4 by OpenAI, capable of processing significantly more words than ChatGPT and even responding to images. Employing an exploratory research design and purposive sampling, the study delves into the responses generated by three AI chatbots: ChatGPT, YouChat, and Chatsonic, revealing issues like response inconsistency, plagiarism from internet sources without proper attribution, and a lack of critical nuances in the generated content, ultimately questioning the credibility and reliability of knowledge dissemination in AELS through these chatbots.[1] The literature review highlights Chat GPT's role in the generative AI revolution, supported by machine learning, deep learning, and natural language processing. It examines current applications in education, noting issues like opacity, data privacy, personalization, and effectiveness. Proposed solutions include explainable algorithms, better encryption, and improved data quality. Future trends point to personalized, intelligent, collaborative, and virtual teaching. Ongoing advancements in related technologies are expected to further enhance generative AI's impact in education and other fields.[2] The paper explores the integration of Large Language Models (LLMs) like ChatGPT in higher education, emphasizing their potential to enhance adaptive learning and teaching approaches. By aligning with the constructivist theory of learning, ChatGPT aids in knowledge construction by summarizing student emails for administrative efficiency and providing personalized responses. While LLMs offer benefits such as research support and automated administrative services, they also present challenges like academic integrity concerns and biases. Educators are advised to balance the advantages of using ChatGPT with the need for ethical and effective implementation in educational

979-8-3315-1795-3/24/$31.00©2024 IEEE

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on October 08,2025 at 02:47:51 UTC from IEEE Xplore. Restrictions apply.

settings. Additionally, the paper highlights the role of ChatGPT in research and writing tasks, serving as a valuable tool for data analysis and writing support, albeit with risks of generating false information. [3] The research paper revises previous recommendations on chatbots and generative AI in scholarly publications to address the growing use of chatbots in scholarly publishing. It underscores concerns regarding the authenticity of content generated by chatbots and aims to guide editors in developing policies for their appropriate use in published papers. The recommendations stress that chatbots cannot be considered authors in scholarly works due to their inability to meet essential authorship criteria. Authors are advised to be transparent about their use of chatbots, acknowledge sources, take public responsibility for their work, and ensure that editors and reviewers are informed about the involvement of chatbots in manuscript evaluation. Furthermore, the paper highlights the necessity for editors to have access to digital tools to effectively manage the impact of chatbots on scholarly publishing, particularly in fields like medical journals where misinformation can have severe repercussions. [4] The paper delves into the potential implications of incorporating generative AIs like ChatGPT in software engineering education, weighing both the challenges and opportunities they present. It raises concerns about the potential limitations of graded homework in this context due to the use of generative AI. However, it also underscores the advantages of leveraging ChatGPT's capabilities to offer personalized feedback to students, thereby enhancing current educational practices in software engineering. The authors advocate for a more individualized approach to education in software engineering facilitated by the availability of generative AI. Furthermore, they stress the need to adapt software engineering curricula to align with the evolving profiles of software engineers in light of integrating generative AI. Additionally, the paper emphasizes the importance of providing guidance on the appropriate utilization of generative AI within courses to prevent any adverse effects on student learning.[5] The research paper delves into the implications of generative AI, particularly focusing on the launch of ChatGPT in November 2022, which evoked mixed reactions among educators. It highlights the application of chatbots responding from large language models (C-LLM) for AI review and assessment of complex student work, showcasing a practical use case for generative AI in educational contexts. The paper also addresses the inherent limitations of generative AI, emphasizing its dependence on language corpora and binary notation for textual representation, which define the boundaries of its capabilities. Despite these constraints, the paper concludes by exploring the emerging and potential applications of generative AI in education, indicating a growing interest in harnessing such technologies for educational purposes.[6] The paper introduces ChatGPT and Generative AI models as transformative technologies that are reshaping AI applications by excelling in human-like conversations and various tasks like question answering and summarization. Despite criticisms for lacking knowledge and reasoning, these models offer sophisticated responses. The challenges highlighted include the impact on academic integrity due to AI-generated content, which can be both beneficial and detrimental in educational and research settings. On the other hand, the paper also discusses the opportunities presented by ChatGPT in enhancing existing chatbots with advanced conversational abilities for intelligent automation across different industries. By exploring the foundational capabilities of ChatGPT and similar models, the paper provides guidelines for effectively leveraging these technologies to address the challenges while maximizing the opportunities they offer in academic and practical contexts.[7] In the realm of education, chatbots have emerged as valuable tools for providing information and support to students, particularly during the university admission process. Prior research has highlighted the efficacy of chatbots in addressing student queries and offering guidance throughout the admission journey. The integration of Natural Language Processing (NLP) techniques in chatbot design plays a pivotal role in enabling these virtual assistants to comprehend and respond to user queries in a more natural and intuitive manner. Leveraging platforms like Dialog-Flow for chatbot development further enhances the conversational capabilities of these systems, allowing for more engaging interactions with users. By harnessing the power of chatbots, universities can streamline the dissemination of information, assist prospective students in navigating admission procedures, exploring study programs, and accessing scholarship details efficiently, ultimately enhancing the overall user experience within university information systems. [8]

III. Dataset

The dataset for this research comprises 250 question-response pairs focused on generating accurate and contextually appropriate answers regarding the Vishwakarma Institute of Technology (VIT) Pune. Each query, covering topics like campus facilities, climate, admissions, security, and academic policies, is transformed into a numerical vector representation. Relevant documents are retrieved from a comprehensive corpus, including official brochures, websites, and student manuals, and are then converted into vector representations to provide the necessary context for response generation. The generative model uses these query and document vectors to produce coherent and accurate responses. The dataset is carefully structured, including query-response pairs and their corresponding document vectors, and evaluated using metrics such as BLEU, ROUGE, and human judgment scores to ensure the quality and relevance of the generated responses. This approach ensures that the model has access to rich and pertinent information, enabling the generation of high-quality responses

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on October 08,2025 at 02:47:51 UTC from IEEE Xplore. Restrictions apply.

IV. PROJECT FLOW DIAGRAM

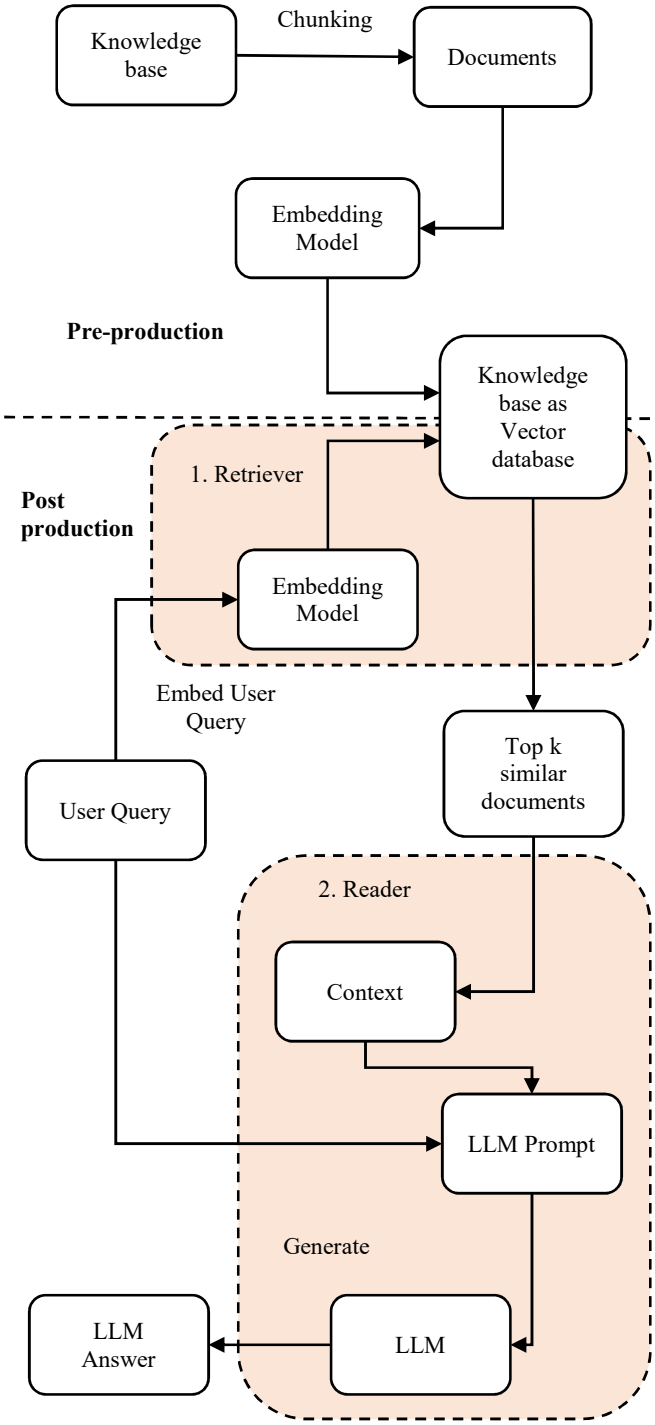

Figure 1: Figure depicting the entire system architecture

V. METHODOLOGY

1. Retrieval-Augmented Generation (RAG) Framework

The Retrieval-Augmented Generation (RAG) framework integrates retrieval and generation techniques to produce accurate and contextually relevant responses. The system architecture comprises several key stages, including document embedding, indexing, query processing, and response generation.

2. System Architecture

The system architecture follows a structured flow to ensure efficient and accurate response generation:

- Document Embedding: University-related documents are transformed into vector embeddings using state-of-the-art language models. This process involves converting textual data into numerical vectors that capture the semantic meaning of the text.
- Indexing: The generated embeddings are indexed into a vector database for efficient retrieval. This step ensures that relevant documents can be quickly accessed based on user queries.
- Query Processing: When a user query is received, the system retrieves relevant documents from the database. The retrieval process identifies the documents most similar to the query in the vector space.
- Response Generation: The retrieved information is used to generate a precise and context-aware response. This step combines the retrieved documents with generative models to formulate an answer.

3. Technologies Used

The following technologies were employed in the development of the chatbot:

- Natural Language Processing (NLP) models for text embedding and generation.
- Vector databases (e.g., FAISS) for efficient document retrieval.
- Machine learning frameworks for integrating the RAG framework.

4. Implementation Details

A. Creating Embeddings

The process of creating vector embeddings from university-related documents involves several steps:

1) Text Extraction: Textual content is extracted from various documents, including PDFs, using text extraction tools. The extracted text is then preprocessed to remove any irrelevant content or noise.

2) Text Splitting: The extracted text is split into smaller chunks to manage the embedding process effectively. This step ensures that the embeddings accurately represent the content and context of the documents.

3) Vector Embedding Generation: Using advanced NLP models, the text chunks are converted into vector embeddings. Mathematically, the embedding process can be represented as:

$$v = f(x) \qquad \ldots(1)$$

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on October 08,2025 at 02:47:51 UTC from IEEE Xplore. Restrictions apply.

where v is the vector embedding and f is the embedding function applied to the text chunk x.

4) Vector Storage: The generated embeddings are stored in a vector database, such as FAISS, allowing for efficient retrieval based on user queries.

B. Main Functionality

1) Model Loading: A language model is loaded to generate responses. The model's parameters are fine-tuned to balance response accuracy and coherence.

2) History-Aware Retrieval: A retriever component takes into account the chat history to formulate standalone questions. This component integrates with the vector database to retrieve relevant documents based on the user's query and chat history.

3) Conversational Chain Setup: A question-answering chain is established using the RAG framework. This chain processes the retrieved documents and generates responses by considering both the content of the documents and the context of the conversation.

4) User Input Handling: The system processes user queries, retrieves relevant information from the vector database, and generates responses using the language model and retriever. The interaction is managed to ensure that responses are accurate and contextually relevant.

C. Mathematical Formulation

1. Document embedding:

   Let $D = (d_1, d_2, d_3 \dots\dots\dots.. d_n)$ be the set of documents. Each document d_i is split into chunks $C_i = (c_{i1}, c_{i2}, \dots\dots\dots.. c_{im})$. The embedding function f maps each chunk c_{ij} to a vector v_{i1}

   $V_{ij} = f(c_{ij})$ ….(2)

2. Query Embedding and Retrieval:

   Given a user query q, the query is embedded into a vector vq. The similarity between the query vector and document vectors is computed using a similarity measure, such as cosine similarity:

   $$(v_q, v_{ij}) = \frac{v_q . v_{ij}}{\|v_q\| \|v_{ij}\|} \quad \dots.(3)$$

   Relevant documents are retrieved based on the highest similarity scores.

3. Response Generation:

   The generative model g takes the retrieved document vectors and the query vector to generate a response r:

   $$r = g\left(v_q, (v_{i1}, v_{i2}, \dots\dots, v_{ik})\right) \quad \dots.(4)$$

VI. Results and Discussion

The precision, recall, and F1-score values obtained from the classification model in the context of crisis response scenarios provide critical insights into its performance. A higher precision signifies the model's accuracy in correctly identifying crisis-related categories, reducing the inclusion of unrelated messages. Concurrently, a higher recall indicates the model's ability to capture a larger proportion of actual crisis-related instances, ensuring comprehensive coverage. The F1-score, balancing precision and recall, offers a consolidated measure of the model's overall effectiveness in categorizing crisis-related messages. Together, these metrics inform stakeholders about the model's capacity to reliably identify and categorize crisis-related information, facilitating timely and appropriate response efforts during emergencies and disasters.

Figure 2 represent the cosine similarity of the generated text by model and the actual text. Achieving a balance between precision, recall, and F1-score is crucial for ensuring the robustness and efficacy of crisis response systems, aiding in the optimization of resource allocation and decision-making processes in crisis management.

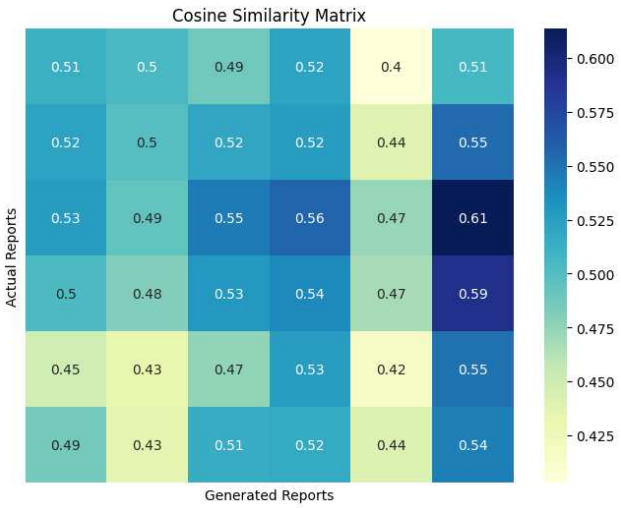

Figure 2: Cosine Similarity matrix of context Retrieval model

VII. Conclusion

This research highlights the successful implementation of a generative AI-driven chatbot using a Retrieval-Augmented Generation framework. The chatbot significantly enhances the user experience by providing precise and context-aware information efficiently. The results indicate that such systems can be invaluable in educational settings, offering a scalable solution to handle numerous inquiries accurately.

References

[1] Chaka, Petrus, Chaka. (2023). Generative AI Chatbots - ChatGPT versus YouChat versus Chatsonic: Use Cases of Selected Areas of Applied English Language Studies. International Journal of Learning, Teaching and Educational Research, 22(6), 1-19. Available from: 10.26803/ijlter.22.6.1

[2] Hao-yan, Yu., Yunyun, Guo. (2023). Generative artificial intelligence empowers educational reform: current status, issues, and prospects. Frontiers in Education, 8 Available from: 10.3389/feduc.2023.1183162

[3] (2023). The role of ChatGPT in higher education: Benefits, challenges, and future research directions. Journal of applied learning and teaching, 6(1) Available from: 10.37074/jalt.2023.6.1.29

[4] Chris, Zielinski., Margaret, Winker., Rakesh, Aggarwal., Lorraine, Ferris., Markus, Heinemann., Sanjay, Pai., Edsel, Ing., Leslie, Citrome., Michael, Voight., Farrokh, Habibzadeh. (2023). Chatbots, Generative AI, and Scholarly Manuscripts: WAME Recommendations on Chatbots and Generative Artificial Intelligence in Relation to

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on October 08,2025 at 02:47:51 UTC from IEEE Xplore. Restrictions apply.

Scholarly Publications. Philippine journal of pathology, 8(1), 6-8. Available from: 10.21141/pjp.2023.08

[5] Marian, Daun., Jennifer, Brings. (2023). How ChatGPT Will Change Software Engineering Education. Available from: 10.1145/3587102.3588815

[6] Anastasia, Olga, Tzirides., Akash, Saini., Gabriela, C., Zapata., Duane, Searsmith., Bill, Cope., Mary, Kalantzis., Jennifer, K., Whiting. (2023). Generative AI: Implications and Applications for Education. arXiv.org, abs/2305.07605 Available from: 10.48550/arXiv.2305.07605

[7] (2023). ChatGPT and Generative AI Guidelines for Addressing Academic Integrity and Augmenting Pre-Existing Chatbots. Available from: 10.1109/icit58465.2023.10143123

[8] Pica, Salsabila, Atmauswan., Akibu, Mahmoud, Abdullahi. (2022). Intelligent Chatbot For University Information System Using Natural Language Approach. 3(2), 59-64. Available from: 10.55862/asbjv3i2a007

[9] (2023). ChatGPT and Generative AI Guidelines for Addressing Academic Integrity and Augmenting Pre-Existing Chatbots. Available from: 10.1109/icit58465.2023.10143123

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on October 08,2025 at 02:47:51 UTC from IEEE Xplore. Restrictions apply.

Authorized licensed use limited to: ANNA UNIVERSITY. Downloaded on October 08,2025 at 02:47:51 UTC from IEEE Xplore. Restrictions apply.